

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

RANDOM PROJECTION

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Johnson-Lindenstrauss Lemma

Summary: **Any set of n points is approximately embeddable in $O(\log n)$ dimensions.**

- Pick any $0 < \epsilon \leq 1/2$ and set $k = (4/\epsilon^2) \log n$.
- Any n points in $\mathbb{R}^d$ can be embedded into $\mathbb{R}^k$, such that each of the interpoint (Euclidean) distances is distorted by at most a multiplicative factor of $1 \pm \epsilon$.
- Moreover, a projection into a random k -dimensional subspace will achieve this with probability close to 1.

How to Project into a Random Subspace?

